

spin-shortener

A polyglot WebAssembly URL shortener built on Fermyon Spin — production target Akamai Functions. Short links, per-link passwords and schedules, QR codes, click analytics, full admin tooling, and a security posture held by tests.

WebAssembly · Spin 4.x

Go + Python

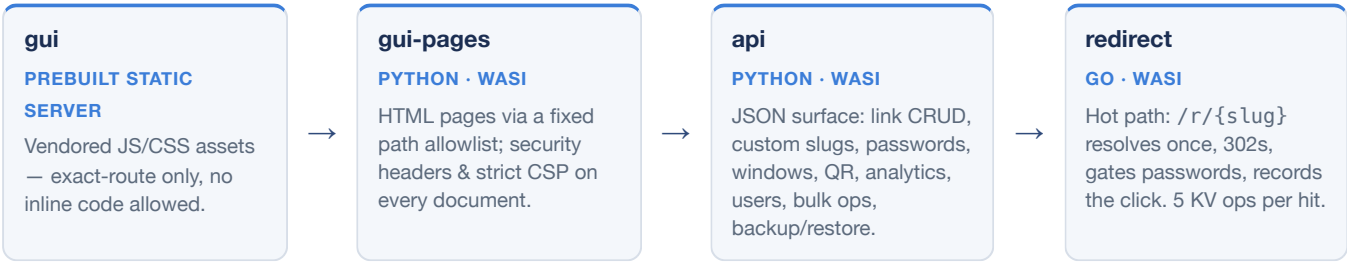
Akamai Functions

Single KV store, 3 namespaces

882 tests, 3 suites

ARCHITECTURE

Four components, six HTTP trigger routes, wired in one `spin.toml`. Go owns the redirect hot path; Python owns authoring, auth and admin — performance where it matters, velocity everywhere else.



CAPABILITIES

- Auto-generated & permission-gated custom short links
- Per-link passwords (PBKDF2, 100k iterations)
- Optional start/end time windows — re-checked on every request
- Per-link QR codes (SVG/PNG), domain-allowlisted
- Click analytics: totals + per-day, 64-way sharded counters
- Bulk create / edit / delete / tag / reassign / repoint / schedule
- Admin-managed destination URL policy (deny-wins rules)
- Multi-domain display; CSV export; link tags & ownership

ADMIN & RELIABILITY

- Local auth, HttpOnly session cookies + CSRF, role/permission model
- User management: create, edit, disable, delete (ownership-gated)
- KV backup / restore — redacted, validated, all-or-nothing
- Consistency check + repair; orphaned-analytics purge
- Write-throttle resilience: bounded retry budgets, best-effort & fully reported bulk writes
- Toggleable structured logging + sanitized failure lines
- X-SS-Version build ID on every response; token-gated Server-Timing tracing
- KV explorer for local development (never deployed)

ENGINEERING HIGHLIGHTS

- ✓ **Polyglot by design** — Go owns the redirect hot path; Python owns authoring/auth/admin.
- ✓ **Single KV store** — one default store, prefixed namespaces, controlled enumeration.
- ✓ **Parallel KV reads** — `get_many` batching, measured against live Akamai quotas.
- ✓ **Security by construction** — no inline code; logging can't carry a KV key; injection closed.
- ✓ **Sharded click counting** — 64 keys/link, cross-language pinned, RAISE-only.
- ✓ **Derived link indexes** — racy RM-W indexes removed; the record is the only truth.
- ✓ **Honest status contract** — 404 product states only; 503/500 faults; 302 never 301/308.
- ✓ **Measurement over assumption** — constants backed by live-deployment measurements.

NUMBERS

4	6	5	64	100k	882	50
COMPONENTS	HTTP TRIGGER ROUTES	KV OPS PER REDIRECT	CLICK SHARDS	PBKDF2 ITERATIONS	TESTS PASSING	WRITE RPS CEILING